

6 Second Application: Monetary Policy Shocks

The previous application involved a panel VAR with the same variable across many different states, lending itself naturally to two-dimensional clustering. We now turn to a less structured setting: a medium-scale VAR of U.S. macroeconomic and financial variables. This allows us to evaluate the method’s performance in high-dimensional systems absent such panel structure and to demonstrate its usefulness for structural analysis. In particular, this application studies impulse responses to monetary policy shocks, where credible interpretation typically depends on the identification strategy.

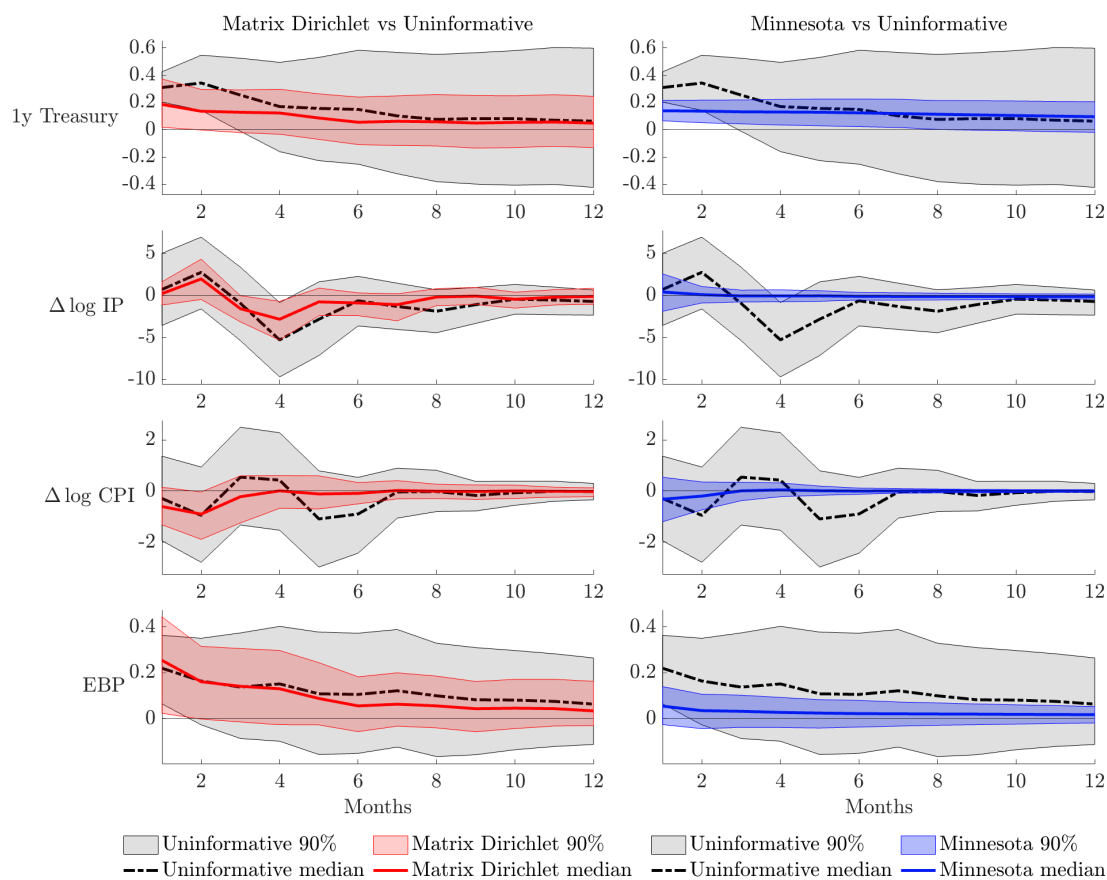
The application follows Plagborg-Møller and Wolf (2021), who show that a proxy variable, ordered first in a Cholesky decomposition, yields impulse responses with a structural interpretation. While large Bayesian VARs have previously been used to study monetary policy (e.g., Bańbura et al., 2010; Crump et al., 2021), they required internal identification schemes, such as recursive orderings or long-run restrictions (Blanchard and Quah, 1989). Recent work has moved toward identification via the use of external instruments, which offer greater empirical credibility (Stock and Watson, 2018), though the applications have so far been limited to small systems, likely in part due to the noisier estimates such an approach produces.⁶ The MDP approach enables us to extend the use of external instruments to high-dimensional VARs, and, in addition, is attractive in connection with the long-standing discussion on the importance of including all variables from the Federal Reserve’s information set (Sims, 1992, Bauer and Swanson, 2023) to avoid omitted variable bias.

We begin by replicating the five-variable VAR analysis in Plagborg-Møller and Wolf (2021), using monthly data on four macroeconomic variables from January 1990 to June 2012 ($T = 270$) and the external instrument for monetary policy shocks constructed by Gertler and Karadi (2015) from high-frequency financial data. The macroeconomic variables — industrial production growth, CPI inflation, the 1-year Treasury yield, and the Excess Bond Premium (EBP) — match those in Gertler and Karadi (2015). Figure 6 presents results from a VAR(4) specification. Under

⁶The use of external instruments with large-dimensional BVARs is especially tricky with Minnesota priors because the instruments can be quite noisy/weak and therefore their coefficients too often get shrunk to zero.

the uninformative prior, the impulse responses replicate the benchmark findings in Plagborg-Møller and Wolf (2021). The Matrix Dirichlet Process prior delivers similar shapes but sharper inference, with the inflation response negative and significant at the 90% level. In contrast, the Minnesota prior, with hyperparameter selected as in Bańbura et al. (2010), shrinks the responses too aggressively, pulling them toward zero.

Figure 6: Comparing impulse responses to a Monetary Policy shock using the Gertler and Karadi (2015) instrument.



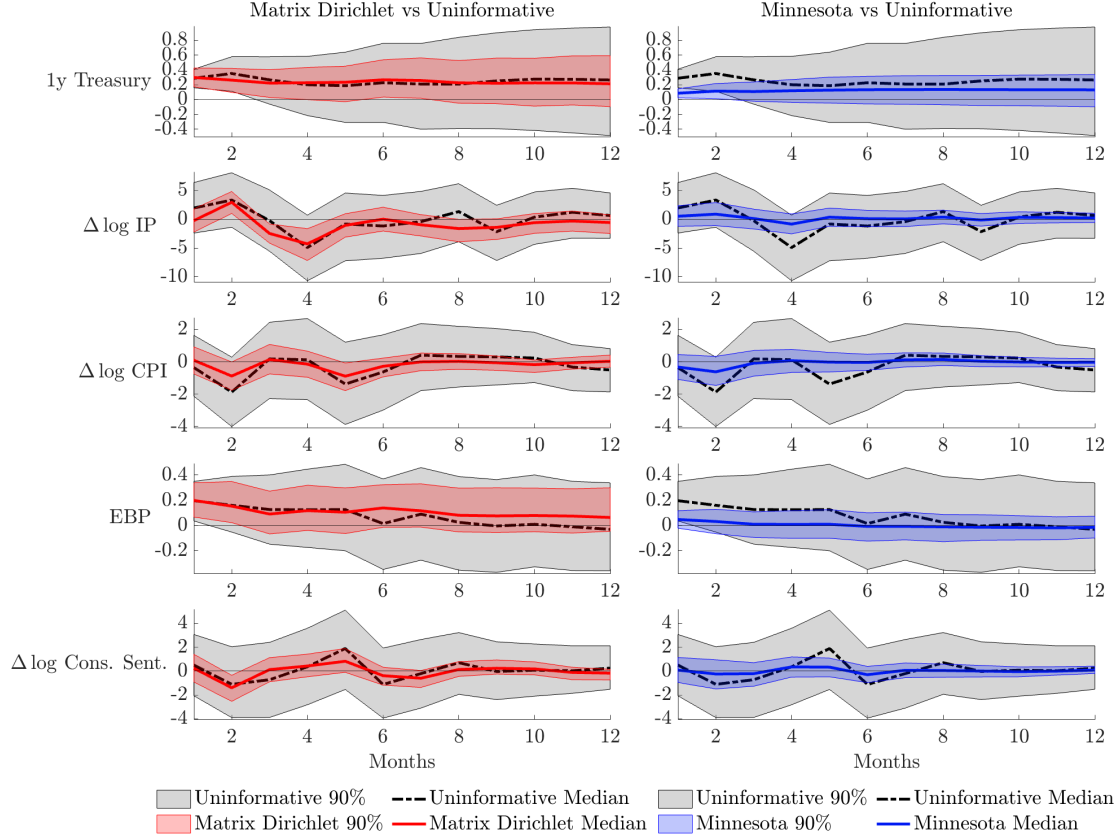
Notes: Impulse response functions from a five-variable VAR(4), compared to using an uninformative prior. The left panel presents IRFs using the Matrix Dirichlet Process prior; the right panel using the Minnesota prior. Shaded areas show 90% credible intervals.

Next, we extend the analysis to an 18-variable VAR(4), using 13 additional time series obtained from the FRED-MD dataset (McCracken and Ng, 2016)⁷. The added variables are a set of real-activity measures (retail sales, building permits, the inventories-to-sales ratio, the civilian unemployment rate), three producer price index (PPI) components (final consumer goods, crude materials, and metals), and a range of financial variables (the BAA–AAA corporate credit spread, the VIX, the S&P 500 index, the S&P dividend yield, the USD/CHF exchange rate, and consumer sentiment). The financial and sentiment variables are forward-looking variables included to control for the potential Federal Reserve information effect (Bauer and Swanson, 2023), and the Producer Price Indices are in response to Sims (1992).

To assess the effect of different priors, Figure 7 compares impulse responses under the uninformative, Matrix Dirichlet Process, and Minnesota priors, focusing on the same four variables as before, i.e., one-year treasury rate, industrial production, CPI inflation, and the excess bond premium, alongside a forward-looking variable: consumer sentiment. For all five variables shown in Figure 7, the uninformative prior yields wide credible intervals that mostly include zero throughout the horizon, except for the excess bond premium at horizon 1. By clustering coefficients on variables with similar dynamic responses, the Matrix Dirichlet Process prior produces narrower intervals that exclude zero at selected horizons, while largely preserving the overall shape and magnitude of the impulse responses. For example, we find significant negative responses in consumer sentiment following an unexpected positive monetary policy shock. In contrast, the Minnesota prior, which shrinks all off-diagonal coefficients toward zero, tends to further attenuate the responses: while intervals are narrower, they uniformly include zero. Figure 8 presents results for a set of financial variables, showing that the Matrix Dirichlet Process prior finds a negative effect on the Swiss-USD exchange rate, while the Minnesota prior again shrinks the effect to zero. Appendix Figure C.1 presents results for the remaining variables, confirming the same patterns.

⁷Data (vintage 2025-05) obtained from <https://www.stlouisfed.org/research/economists/mccracken/fred-databases> on June 24, 2025.

Figure 7: Selected impulse responses to a monetary policy shock in an 18-variable VAR(4) model, comparing uninformative prior to the Matrix Dirichlet Process prior (left, red) and the Minnesota prior (right, blue)



7 Third Application: Uncertainty Shocks and PPI Spillovers

The macroeconomic effects of uncertainty shocks have been analyzed by, among others, Bloom (2009), Carriero, Mumtaz, Theodoridis, and Theophilopoulou (2015), Leduc and Liu (2016), Basu and Bundick (2017), and Fasani and Rossi (2018). Empirical evidence indicates that increased uncertainty – often proxied by the volatility index (VIX) – tends to lower prices, typically identified by placing the VIX or another volatility proxy first in the Cholesky ordering of a VAR. In this application, we wish to analyze how uncertainty shocks propagate, with particular emphasis on the channels through which these shocks affect sector-level prices across the supply-chain network.